AMAN TIWARI

GEOSPATIAL SOLUTIONS | GIS & REMOTE SENSING | TECHNICAL CONSULTING

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PROFESSIONAL SUMMARY

GIS and Remote Sensing professional with 4+ years of experience designing and delivering geospatial solutions using satellite, drone and field data. Led the geospatial workstream in building a carbon dMRV platform from the ground up, from interpreting business requirements and designing spatial data structures to analytical workflows and internal implementation. Experience spans infrastructure monitoring, climate-risk assessment, land-use analytics and cross-functional technical delivery.

PROFESSIONAL EXPERIENCE

Product & GIS Manager | FCF India, New Delhi | Feb 2025 - Present

- Led the geospatial workstream for a carbon digital Monitoring, Reporting and Verification (dMRV) platform from initial requirements to internal use, translating ARR and nature-based solutions monitoring needs into technical priorities.
- Designed and initially implemented GCP-based geospatial processing workflows and the PostGIS database schema; handed off system designs to engineers for further scaling and refinement.
- Built Python and Google Earth Engine pipelines for vegetation-index time series (NDVI, SAVI, MSAVI), land-cover classification, change detection and parcel-level carbon monitoring.
- Integrated satellite imagery, drone data and GPS/field surveys with spatial QA/QC, maps and technical documentation supporting monitoring and verification readiness.
- Implemented a drone-imagery tiling workflow that reduced load time from approximately 2-3 minutes to a few seconds; led a team of three GIS analysts.

GIS Engineer | DronaMaps, Gurugram | Apr 2024 - Feb 2025

- Led GIS workflows within the NHAI Drone Analytics Monitoring System (DAMS), supporting multi-temporal mapping and change detection across an 11,000+ km national highway monitoring program.
- Developed and deployed interactive Esri Web AppBuilder dashboards supporting internal leadership, NHAI and partner stakeholders in infrastructure monitoring and decision-making.
- Directly managed 35 GIS interns, overseeing task allocation, geospatial data-processing workflows, quality assurance and delivery.

GIS Consultant | CEEW, New Delhi | Sep 2023 - Apr 2024

- Used Google Earth Engine and Python for spatial and temporal heat-stress analysis contributing to the Heat Action Plan for Thane.
- Helped design the Climate Resilience Atlas for India, bringing together 15+ geospatial layers for district-level climate-risk analysis and decision support.

EARLIER EXPERIENCE

GIS Executive | Attentive.ai (Remote, Mar-Sep 2021): Satellite/aerial imagery feature mapping and GIS data QA for automated extraction workflows.

GIS Consultant & Trainer | AGSRT (Remote, Sep 2020-Mar 2021): GIS training in QGIS and ArcGIS, spatial analysis and survey-data workflows.

GIS Data Intern | Intensel (Remote, Jun 2020): Prepared satellite-based climate and flood-risk geospatial datasets.

TECHNICAL SKILLS

GIS & Remote Sensing: Google Earth Engine (Python/JavaScript), QGIS, ArcGIS Pro, GeoServer, satellite and drone imagery, raster/vector analysis, LULC, vegetation indices, change detection.

Programming & Data: Python, Pandas, NumPy, GeoPandas, Rasterio, GDAL, SQL, PostGIS, spatial databases and workflow automation.

Geospatial Systems & Applications: GCP (Cloud Run, Cloud Storage), spatial data integration, web GIS and dashboards, GIS QA/QC, carbon MRV/VM0047, climate-risk and infrastructure monitoring.

EDUCATION

M.Sc. Geo-Information Science & Earth Observation | University of Twente (ITC), Netherlands (2021-2023)

B.Tech. Geoscience Engineering | University of Petroleum & Energy Studies (UPES), Dehradun (2016-2020)

SELECTED ACADEMIC PROJECT

3D Flood Visualization Web GIS (M.Sc. thesis) - Developed a web-based 3D visualization prototype using prepared terrain and geospatial datasets to explore urban flood scenarios (CesiumJS, Three.js, Python).